

Verification of the CSMA/CA Bianchi Interference Model in the `ncsim` Networked Compute Simulator

Autonomous Networks Research Group (ANRG)
University of Southern California

March 2026

Abstract

We present a systematic verification of the `csma_bianchi` wireless interference model implemented in `ncsim`, a headless discrete-event simulator for networked computing research. The model separates two distinct 802.11 interference mechanisms: (1) CSMA/CA contention-domain time-sharing modeled by Bianchi’s saturation throughput analysis, and (2) hidden-terminal SINR degradation from links outside the carrier-sensing range. Hidden-terminal interference uses an *expected-rate* model that weights each hidden terminal’s contribution by its Bianchi transmit probability, rather than assuming worst-case simultaneous transmission. Nine experiments compare simulator output against closed-form analytical predictions across a range of topologies, from single-link baselines to five-link cascading hidden-terminal scenarios and combined bandwidth-sharing-plus-interference configurations. All 113 individual test points match predictions within a 1% tolerance, validating the correctness of the interference model, the execution engine’s dynamic bandwidth recalculation, and the per-link fair-sharing mechanism.

1 Introduction

Accurate modeling of wireless link interference is essential for any networked-computing simulator that aims to produce realistic transfer-time estimates. In multi-hop or multi-link wireless networks, concurrent transmissions interact through two fundamentally different mechanisms:

1. **Contention-domain time-sharing.** Links whose transmitters can sense each other (i.e., within carrier-sensing range) contend for channel access via the 802.11 CSMA/CA protocol. These links share airtime but do not transmit simultaneously, so they do not cause signal-level interference at each other’s receivers.
2. **Hidden-terminal SINR degradation.** Links whose transmitters *cannot* sense each other may transmit simultaneously, causing co-channel interference at the receiver. The resulting signal-to-interference-plus-noise ratio (SINR) determines the achievable MCS and data rate.

The `ncsim` simulator implements these two mechanisms in its `csma_bianchi` interference model. The combined effective data rate for a link is:

$$R_{\text{eff}} = \frac{R_{\text{base}} \cdot f_{\text{SINR}} \cdot \eta(n)}{n \cdot N_{\text{flows}}} \quad (1)$$

where R_{base} is the SNR-only PHY rate, f_{SINR} captures hidden-terminal degradation, $\eta(n)/n$ is the Bianchi per-station throughput share among n contending links, and N_{flows} is the number of concurrent data transfers sharing the same physical link.

This report describes nine experiments that verify Equation (1) and the simulator’s dynamic recalculation behavior against closed-form predictions. Section 2 describes the interference model in detail. Sections 3–11 present each experiment. Section 12 summarizes the results.

2 System Model

2.1 RF Parameters

All experiments use the default `RFConfig` parameters listed in Table 1.

Table 1: Default RF configuration.

Parameter	Value	Unit
Transmit power (P_{tx})	20.0	dBm
Carrier frequency (f)	5.0	GHz
Path-loss exponent (n_{PL})	3.0	–
Noise floor (N_0)	−95.0	dBm
CCA threshold	−82.0	dBm
Channel width	20	MHz
WiFi standard	802.11ax	–

2.2 Path Loss and Rate Selection

The log-distance path-loss model gives the received power at distance d :

$$P_{\text{rx}}(d) = P_{\text{tx}} - \text{PL}(d_0) - 10 n_{\text{PL}} \log_{10}(d/d_0) \quad (2)$$

where $\text{PL}(d_0) = 20 \log_{10}(4\pi d_0 f/c)$ is the Friis free-space loss at reference distance $d_0 = 1$ m. The signal-to-noise ratio is $\text{SNR} = P_{\text{rx}} - N_0$ (in dB). The highest 802.11ax MCS whose minimum SNR requirement is satisfied determines the PHY data rate.

2.3 Carrier Sensing and Conflict Graph

The carrier-sensing range is the maximum distance at which a transmission triggers the CCA:

$$d_{\text{CS}} = d_0 \cdot 10^{(P_{\text{tx}} - \text{CCA} - \text{PL}(d_0))/(10 n_{\text{PL}})} \quad (3)$$

With the default parameters, $d_{\text{CS}} = 71.2$ m. Two links *conflict* (without RTS/CTS) if the transmitter of either link can sense any node of the other link. The set of all pairwise conflicts forms the *conflict graph*.

2.4 Bianchi MAC Efficiency

For n stations contending under 802.11 DCF, Bianchi’s saturation-throughput analysis [1] yields a MAC efficiency $\eta(n) \in (0, 1]$ representing the fraction of channel time occupied by successful payload delivery. Each of the n contending links obtains a throughput share of $\eta(n)/n$. For $n = 1$, $\eta(1) = 1.0$ (no contention overhead).

2.5 Hidden-Terminal SINR (Expected-Rate Model)

Active links *not* in the conflict graph of a given link are its hidden terminals. Unlike contention-domain neighbors, hidden terminals are not coordinated by CSMA and may transmit simultaneously with the victim link. However, each hidden terminal h does not transmit 100% of the time—it shares its own contention domain with its own conflict-graph neighbors. Let n_h be the number of active links in h ’s contention domain (including h itself). Then h transmits with probability:

$$p_h = \frac{\eta(n_h)}{n_h} \quad (4)$$

Rather than assuming all hidden terminals transmit simultaneously (worst case), the model computes an *expected rate* by enumerating all 2^N subsets of the N hidden terminals. For each subset $S \subseteq \mathcal{H}$, the SINR is computed from only the interferers in S :

$$\text{SINR}(S) = \frac{P_{\text{signal}}}{N_0 + \sum_{j \in S} P_j} \quad (5)$$

The probability of subset S is $\Pr(S) = \prod_{j \in S} p_j \prod_{j \notin S} (1 - p_j)$, and each subset yields a rate $R(S)$ via MCS selection. The expected rate is:

$$R_{\text{exp}} = \sum_{S \subseteq \mathcal{H}} \Pr(S) \cdot R(S) \quad (6)$$

The SINR factor is then $f_{\text{SINR}} = R_{\text{exp}}/R_{\text{base}}$. When $N > 10$ (rare in practice), the model falls back to the worst-case $S = \mathcal{H}$.

Note that when all hidden terminals have $p_h = 1.0$ (e.g., each is the sole link in its contention domain), every subset has zero probability except the full set $S = \mathcal{H}$, and the expected-rate model reduces to the simpler worst-case SINR formula.

2.6 Combined Interference Factor

The `csma_bianchi` model composes the two mechanisms multiplicatively:

$$f = f_{\text{SINR}} \cdot \frac{\eta(n)}{n}, \quad f \in [0.01, 1.0] \quad (7)$$

The effective per-flow bandwidth on a link carrying N_{flows} concurrent transfers is then $R_{\text{base}} \cdot f/N_{\text{flows}}$.

2.7 Dynamic Recalculation

When any transfer starts or completes, *all* other active links are recalculated. Partial progress is tracked via a `data_remaining` field:

$$D'_{\text{rem}} = D_{\text{rem}} - R_{\text{eff}} \cdot \Delta t \quad (8)$$

where Δt is the elapsed time since the last recalculation. A new completion time is then scheduled based on D'_{rem} and the updated effective rate.

2.8 Verification Methodology

Each experiment generates a scenario YAML, runs it through `ncsim`, and compares simulated transfer durations (or effective rates) against closed-form analytical predictions computed from the same RF functions. A match is declared when the relative error is within 1%.

3 Experiment 1: Link Length vs. Data Rate

3.1 Setup

A single 802.11ax link connects two nodes at varying distances $d \in \{1, 3, 5, 8, \dots, 140\}$ m. A single 10 MB transfer is sent across the link with no other active links (no interference). The predicted rate is simply the SNR-only PHY rate $R_{\text{base}}(d)$.

3.2 Purpose

Validates the baseline PHY-layer pipeline: distance $\rightarrow$ path loss $\rightarrow$ SNR $\rightarrow$ MCS selection $\rightarrow$ data rate. This is a prerequisite for all subsequent experiments.

3.3 Results

Table 2: Experiment 1: Single-link PHY rate vs. distance.

d (m)	PL (dB)	SNR (dB)	MCS	Predicted (MB/s)	Simulated (MB/s)	Match
1	46.42	68.58	11	17.9250	17.9250	✓
5	67.39	47.61	11	17.9250	17.9250	✓
12	78.80	36.20	9	14.3375	14.3375	✓
20	85.45	29.55	7	10.7500	10.7500	✓
30	90.73	24.27	5	8.6000	8.6000	✓
50	97.39	17.61	3	4.3000	4.3000	✓
75	102.67	12.33	2	3.2250	3.2250	✓
95	105.75	9.25	1	2.1500	2.1500	✓
120	108.80	6.20	0	1.0750	1.0750	✓
140	110.81	4.19	n/a	0.0000	0.0000	✓

All 20 test points match exactly. The rate decreases in discrete steps corresponding to MCS thresholds, and drops to zero beyond $d = 130$ m where the SNR falls below the minimum MCS 0 requirement of 5 dB.

4 Experiment 2: Parallel Link Separation

4.1 Setup

Two parallel horizontal links of length 30 m are placed at vertical separation $s \in \{5, 10, \dots, 200\}$ m. Link A runs from $n_0(0, 0)$ to $n_1(30, 0)$; Link B from $n_2(0, s)$ to $n_3(30, s)$. Each carries a 10 MB transfer. When $s \leq d_{CS} = 71.2$ m, both links are in the conflict graph (Bianchi contention). When $s > d_{CS}$, they are hidden terminals (SINR degradation).

4.2 Predictions

- **Conflict regime** ($s \leq 71.2$ m): $R = R_{\text{base}} \cdot \eta(2)/2 = 8.6 \times 0.4404 = 3.7871$ MB/s.
- **Hidden-terminal regime** ($s > 71.2$ m): The interferer distance is $d_I = \sqrt{30^2 + s^2}$. Both links see the same interferer geometry, so rates are symmetric: $R = R_{\text{SINR}}(30, d_I)$.

4.3 Results

Table 3: Experiment 2: Two parallel links at varying separation.

s (m)	In Conflict	Pred. R_A	Sim. R_A	Pred. R_B	Sim. R_B
5	Yes	3.7871	3.7871	3.7871	3.7871
20	Yes	3.7871	3.7871	3.7871	3.7871
50	Yes	3.7871	3.7871	3.7871	3.7871
70	Yes	3.7871	3.7871	3.7871	3.7871
75	No	3.2250	3.2250	3.2250	3.2250
80	No	3.2250	3.2250	3.2250	3.2250
100	No	4.3000	4.3000	4.3000	4.3000
140	No	6.4500	6.4500	6.4500	6.4500
200	No	6.4500	6.4500	6.4500	6.4500

All 15 test points match. In the conflict regime the rate is constant at $\eta(2)/2 \approx 0.44$ of the base rate regardless of separation. In the hidden-terminal regime, the rate initially drops (at $s = 75$ m the SINR degrades the MCS) then recovers as the interferer becomes more distant. Both links always receive the same rate, confirming symmetric behavior.

5 Experiment 3: Two Transmitters to Same Receiver

5.1 Setup

Two transmitter nodes $n_1(d, 0)$ and $n_2(0, d)$ both send 10 MB to a shared receiver $n_0(0, 0)$, positioned at 90° apart at equal distance d . This tests the model when two links share a receiver, for $d \in \{5, 10, \dots, 130\}$ m.

5.2 Predictions

In the conflict regime ($d \leq 71.2$ m, since each transmitter can sense n_0 at distance d), the rate is $R_{\text{base}}(d) \cdot \eta(2)/2$. In the hidden-terminal regime, both interferers are at distance d from the shared receiver, causing equal-power interference. At $d = 75$ m this yields a SINR so low that the model clamps to the minimum factor of 0.01.

5.3 Results

Table 4: Experiment 3: Two transmitters to shared receiver.

d (m)	In Conflict	Pred. R_{l10}	Sim. R_{l10}	Pred. R_{l20}	Sim. R_{l20}
5	Yes	7.8934	7.8934	7.8934	7.8934
20	Yes	4.7338	4.7338	4.7338	4.7338
30	Yes	3.7871	3.7871	3.7871	3.7871
50	Yes	1.8935	1.8935	1.8935	1.8935
70	Yes	1.4201	1.4201	1.4201	1.4201
75	No	0.0323	0.0323	0.0323	0.0323
100	No	0.0215	0.0215	0.0215	0.0215
130	No	0.0108	0.0108	0.0108	0.0108

All 18 test points match. The dramatic rate collapse at $d = 75$ m occurs because equal-power hidden-terminal interference at the shared receiver yields an extremely low SINR, demonstrating the model’s minimum-factor clamping behavior.

6 Experiment 4: Three Parallel Links

6.1 Setup

Three parallel 30 m links are placed at $y = 0$, $y = s$, and $y = 2s$, labeled A (outer), B (middle), and C (outer). Each carries a 10 MB transfer. As separation s increases, three interference regimes emerge:

- **All-conflict** ($s \leq 35.6$ m): All three pairs are within d_{CS} . Each link gets $R_{\text{base}} \cdot \eta(3)/3$.
- **Mixed** ($35.6 < s \leq 71.2$ m): Adjacent pairs (A–B, B–C) are in conflict, but the outer pair (A–C, at distance $2s > d_{\text{CS}}$) are hidden terminals. The expected-rate model (Equation 6) applies to the SINR factor.

- **All-hidden** ($s > 71.2\text{ m}$): No conflict-graph edges. Each link sees the others purely as hidden terminals with multi-interferer SINR.

6.2 Multi-Phase Prediction

Because A/C and B generally have different rates, one group finishes before the other. When links complete, the remaining links' interference conditions change, creating a second phase with updated rates. The analytical prediction accounts for this by computing:

1. Phase 1 rates from the initial interference conditions.
2. Which group finishes first (smallest D/R).
3. Phase 2 rates for remaining links with reduced interference.
4. Effective rate = $D/(\text{Phase 1 duration} + \text{Phase 2 duration})$.

6.3 Mixed-Regime Analysis

In the mixed regime, the expected-rate model produces a physically meaningful ordering where outer links A and C achieve *higher* throughput than middle link B. The analysis is as follows:

Outer links (A and C). Each contends with B in a 2-way Bianchi domain, receiving an airtime share of $\eta(2)/2$. Each also has one hidden terminal—the far outer link—at interferer distance $d_I^{\text{far}} = \sqrt{30^2 + (2s)^2}$. Crucially, this hidden terminal also contends with B in its own 2-way domain, so its transmit probability is $p_h = \eta(2)/2 \approx 0.44$, not 1.0. By Equation (6), the expected SINR rate is:

$$R_{\text{exp}}^A = p_h \cdot R_{\text{SINR}}(d_I^{\text{far}}) + (1 - p_h) \cdot R_{\text{base}} \quad (9)$$

Since $p_h < 1$, more than half the time the far hidden terminal is *not transmitting*, and A operates at the full base rate. This significantly boosts A's expected rate compared to the worst-case model.

Middle link (B). B contends with both A and C in a 3-way Bianchi domain, receiving a smaller airtime share of $\eta(3)/3 \approx 0.286$. B has no hidden terminals (both neighbors are in its conflict graph), so $f_{\text{SINR}} = 1.0$. Despite having no SINR degradation, B's rate is limited by the heavier 3-way contention.

Result. The combination of lighter contention ($\eta(2)/2 > \eta(3)/3$) and probability-weighted SINR (most of the time, no interference) gives A and C a higher effective rate than B.

6.4 Drop at the Mixed-to-All-Hidden Boundary

An apparent paradox occurs at the regime transition near $s = 71.2\text{ m}$: throughput *decreases* when links move from the mixed to the all-hidden regime. At $s = 70\text{ m}$ (mixed), $R_A = 3.37\text{ MB/s}$; at $s = 75\text{ m}$ (all-hidden), $R_A = 3.23\text{ MB/s}$.

The explanation lies in the loss of CSMA protection. In the mixed regime, A's near neighbor B is in the conflict graph, meaning CSMA prevents A and B from transmitting simultaneously—B cannot cause SINR degradation at A's receiver. When the separation exceeds d_{CS} and B becomes a hidden terminal instead, B now transmits with $p = 1.0$ (no conflict neighbors) simultaneously with A, causing SINR degradation from a close-range interferer at distance $\sqrt{30^2 + s^2} \approx 81\text{ m}$.

The SINR degradation from this newly “exposed” near interferer is more harmful than the airtime-sharing penalty that CSMA contention had imposed. In wireless networking terms, hidden-terminal collisions are worse than contention-domain time-sharing when the interferer is close enough to significantly degrade the signal quality.

6.5 Convergence to Solo Rate

As s increases further, interference power from hidden terminals decreases with distance. At $s = 300$ m, the nearest interferer is at distance $\sqrt{30^2 + 300^2} \approx 301.5$ m. At this distance, the received interference power is approximately -101 dBm, which is 6 dB below the noise floor of -95 dBm. The interference becomes negligible and all three links converge to the solo rate of $R_{\text{base}} = 8.6$ MB/s.

The convergence is asymmetric: outer links A and C reach the solo rate at $s = 250$ m (their far interferer at $\sqrt{30^2 + (2 \times 250)^2} \approx 501$ m is negligible, and the near one at ~ 251 m is also below the noise floor). Middle link B, with *two* near interferers at the same distance, requires $s = 300$ m to become fully independent.

6.6 Results

Table 5: Experiment 4: Three parallel links at varying separation.

s (m)	Regime	Pred R_A	Sim R_A	Pred R_B	Sim R_B	Pred R_C	Sim R_C	A=C
10	all_conf	2.4611	2.4611	2.4611	2.4611	2.4611	2.4611	✓
20	all_conf	2.4611	2.4611	2.4611	2.4611	2.4611	2.4611	✓
35	all_conf	2.4611	2.4611	2.4611	2.4611	2.4611	2.4611	✓
40	mixed	2.7448	2.7448	2.6571	2.6571	2.7448	2.7448	✓
50	mixed	2.9532	2.9532	2.7934	2.7934	2.9532	2.9532	✓
60	mixed	2.9532	2.9532	2.7934	2.7934	2.9532	2.9532	✓
70	mixed	3.3701	3.3701	3.0480	3.0480	3.3701	3.3701	✓
75	all_hid	3.2250	3.2250	2.8667	2.8667	3.2250	3.2250	✓
80	all_hid	3.2250	3.2250	2.8667	2.8667	3.2250	3.2250	✓
100	all_hid	4.3000	4.3000	3.8222	3.8222	4.3000	4.3000	✓
150	all_hid	6.4500	6.4500	5.1600	5.1600	6.4500	6.4500	✓
200	all_hid	6.4500	6.4500	6.4500	6.4500	6.4500	6.4500	✓
250	all_hid	8.6000	8.6000	6.8800	6.8800	8.6000	8.6000	✓
300	all_hid	8.6000	8.6000	8.6000	8.6000	8.6000	8.6000	✓
400	all_hid	8.6000	8.6000	8.6000	8.6000	8.6000	8.6000	✓
500	all_hid	8.6000	8.6000	8.6000	8.6000	8.6000	8.6000	✓

All 16 test points match. The outer links A and C always have identical rates (geometric symmetry), confirming symmetric recalculation. The results demonstrate three noteworthy behaviors:

1. In the mixed regime ($s = 40\text{--}70$ m), A/C are consistently *faster* than B, due to lighter contention and probability-weighted SINR (Section 4.3).
2. At the mixed-to-all-hidden boundary ($s = 70 \rightarrow 75$ m), throughput drops as CSMA protection is lost and near-neighbor collisions dominate (Section 4.4).
3. All links converge to the solo rate of 8.6 MB/s by $s = 300$ m, with outer links converging first at $s = 250$ m (Section 4.5).

7 Experiment 5: Staggered Transfer Start

7.1 Setup

Two parallel 30 m links at 100 m separation (hidden terminals). Link B starts transferring immediately, while Link A’s transfer is delayed by δ seconds (its predecessor task has a large compute cost). This tests the simulator’s ability to dynamically recalculate in-flight transfers when interference conditions change mid-transfer.

7.2 Predictions

Let $R_0 = 8.6$ MB/s (solo rate) and $R_1 = 4.3$ MB/s (both-active SINR rate). Three phases occur:

1. Phase 1: B transmits solo at R_0 for δ seconds.
2. Phase 2: Both active at R_1 until B finishes.
3. Phase 3: A transmits solo at R_0 until A finishes.

By symmetry, both transfers have the same total duration: $T = \delta + (D - \delta R_0)/R_1$.

7.3 Results

Table 6: Experiment 5: Staggered start with hidden terminals.

Delay Cost	δ (s)	Pred. Duration	Dur. A	Dur. B	A=B
10000	0.1000	2.225591	2.225591	2.225591	✓
30000	0.3000	2.025591	2.025591	2.025591	✓
50000	0.5000	1.825591	1.825591	1.825591	✓
80000	0.8000	1.525591	1.525591	1.525591	✓

All 4 test points match exactly. The A=B symmetry confirms that when A starts and B is recalculated, B’s rate degradation exactly mirrors A’s, and when B finishes, A correctly recovers to the solo rate.

8 Experiment 6: Per-Link Bandwidth Sharing

8.1 Setup

Multiple concurrent flows (3 or 4) share a single 30 m link with no inter-link interference. Each flow carries a different data size. Source tasks are queued on the same node (FIFO), creating a negligible $\sim 10 \mu\text{s}$ stagger between transfer starts.

This experiment isolates the execution engine’s per-link fair-sharing mechanism: N flows on one link each receive R_{base}/N of the bandwidth. When a flow completes, remaining flows are recalculated with a larger share.

8.2 Multi-Phase Prediction

With data sizes $D_1 < D_2 < \dots < D_N$ and link bandwidth B :

- Phase 1: All N flows at B/N . The smallest (D_1) finishes at $t_1 = D_1 \cdot N/B$.
- Phase 2: $N-1$ flows at $B/(N-1)$. The next-smallest finishes after its remaining data is transferred.
- Continue until the last flow completes solo at rate B .

8.3 Results

Table 7: Experiment 6: Per-link bandwidth sharing with multiple flows.

Test Case	Flow	Data (MB)	Pred. Duration (s)	Sim. Duration (s)
3-flow	Ts0	5.0	1.744186	1.744161
3-flow	Ts1	10.0	2.906977	2.906962
3-flow	Ts2	15.0	3.488372	3.488352
3-flow-eq	Ts0	5.0	1.744186	1.744161
3-flow-eq	Ts1	5.0	1.744186	1.744171
3-flow-eq	Ts2	15.0	2.906977	2.906957
4-flow	Ts0	3.0	1.395349	1.395306
4-flow	Ts1	6.0	2.441860	2.441837
4-flow	Ts2	9.0	3.139535	3.139512
4-flow	Ts3	12.0	3.488372	3.488342

All 10 test points match within tolerance. The “3-flow-eq” case validates that two flows with equal data sizes complete at essentially the same time (within microseconds), and the remaining flow correctly transitions to a higher rate. The “4-flow” case exercises four sequential phase transitions as flows complete one by one.

9 Experiment 7: N -Way Bianchi Contention Scaling

9.1 Setup

N parallel 30 m links are placed at 5 m vertical separation, all within carrier-sensing range of each other. This creates a complete conflict graph of size N . Each link carries a 10 MB transfer. Since all links have the same rate, they form a single-phase scenario.

9.2 Predictions

With all N links in a complete conflict graph, the factor for each link is $\eta(N)/N$, giving $R = R_{\text{base}} \cdot \eta(N)/N = 8.6 \cdot \eta(N)/N$.

9.3 Results

Table 8: Experiment 7: Bianchi contention scaling with N links.

N	$\eta(N)$	$\eta(N)/N$	Pred. Rate (MB/s)	Sim. Rate (MB/s)	Match
2	0.8807	0.4404	3.7871	3.7871	✓
3	0.8585	0.2862	2.4611	2.4611	✓
4	0.8367	0.2092	1.7989	1.7989	✓
5	0.8179	0.1636	1.4067	1.4067	✓
6	0.8019	0.1336	1.1494	1.1494	✓
7	0.7882	0.1126	0.9683	0.9683	✓
8	0.7257	0.0907	0.7802	0.7802	✓

All 7 test points match. The MAC efficiency $\eta(N)$ decreases gracefully from 0.88 at $N=2$ to 0.73 at $N=8$, while the per-link share $\eta(N)/N$ decreases much faster due to the $1/N$ divi-

sion. All N links within each test case receive identical rates, confirming symmetric contention behavior.

10 Experiment 8: Five-Link Hidden-Terminal Cascade

10.1 Setup

Five parallel 30 m links at 100 m vertical separation (all pairs beyond d_{CS} , hence all hidden terminals). Data sizes are $\{2, 4, 6, 8, 10\}$ MB to force links to complete at different times, creating up to four cascading phase transitions. The five links have asymmetric interference geometry: outer links (at $y = 0$ and $y = 400$) have one near and three increasingly distant interferers, while the middle link (at $y = 200$) has two near and two far interferers.

10.2 Multi-Phase Prediction

A general cascade prediction is used:

1. Compute SINR-based rates for all active links given current interferer set.
2. Advance time until the next link completes.
3. Remove completed link; recompute rates with fewer interferers.
4. Repeat until all links finish.

This produces up to 5 phases with successively fewer active links.

10.3 Results

Table 9: Experiment 8: Five-link hidden-terminal cascade.

Link	Position y	Data (MB)	Pred. Duration (s)	Sim. Duration (s)	Match
l_0	0	2.0	0.465116	0.465116	✓
l_1	100	4.0	1.046512	1.046512	✓
l_2	200	6.0	1.656977	1.656977	✓
l_3	300	8.0	2.274709	2.274709	✓
l_4	400	10.0	2.300145	2.300145	✓

All 5 test points match. This experiment is the most demanding stress test: five links undergo four phase transitions, each requiring correct `data_remaining` tracking and SINR re-computation with a changing active-link set. Notably, links l_3 and l_4 finish at nearly the same time despite having different data sizes (8 vs. 10 MB), because l_4 (an outer link) has a higher rate than l_3 (an inner link) in all phases.

11 Experiment 9: Combined Bandwidth Sharing and Interference

11.1 Setup

This experiment combines both mechanisms: two flows share link l_{01} ($n_0 \rightarrow n_1$, 30 m) while a third flow on link l_{23} ($n_2 \rightarrow n_3$, 30 m at 100 m separation) acts as a hidden terminal. The effective

per-flow rate on l_{01} is therefore:

$$R_{\text{flow}} = \frac{R_{\text{base}} \cdot f_{\text{SINR}}}{N_{\text{flows}}} = \frac{8.6 \times 0.5}{2} = 2.15 \text{ MB/s} \quad (10)$$

where $f_{\text{SINR}} = R_{\text{SINR}}(30, 104.4)/R_{\text{base}} = 4.3/8.6 = 0.5$ accounts for the hidden terminal, and $N_{\text{flows}} = 2$ accounts for per-link fair sharing. Three test cases exercise different phase orderings by varying data sizes.

11.2 Phase Analysis (Case 1: l_{01} : 5/10 MB, l_{23} : 8 MB)

- **Phase 1** (all active): l_{01} flows at 2.15 MB/s each; l_{23} at 4.3 MB/s. The l_{23} flow (8 MB) finishes first at $8/4.3 = 1.86$ s.
- **Phase 2** (l_{01} only, 2 flows, no interference): Each flow at $8.6/2 = 4.3$ MB/s. The 5 MB flow finishes.
- **Phase 3** (l_{01} only, 1 flow): Solo at 8.6 MB/s. The 10 MB flow finishes.

11.3 Results

Table 10: Experiment 9: Combined bandwidth sharing and hidden-terminal interference.

Case	Flow	Data (MB)	Link	Pred. Duration (s)	Sim. Duration (s)
1	Ta0	5.0	l_{01}	2.093023	2.093018
1	Tb0	10.0	l_{01}	2.674419	2.674408
1	Tc0	8.0	l_{23}	1.860465	1.860465
2	Ta0	5.0	l_{01}	1.511628	1.511623
2	Tb0	10.0	l_{01}	2.093023	2.093013
2	Tc0	3.0	l_{23}	0.697674	0.697674
3	Ta0	3.0	l_{01}	1.395349	1.395339
3	Tb0	15.0	l_{01}	4.186047	4.186037
3	Tc0	20.0	l_{23}	4.418605	4.418605

All 9 test points match. Case 1 has the l_{23} flow finishing first, removing interference before the l_{01} sharing phases complete. Case 2 has l_{23} finishing very early, so most of the l_{01} transfer occurs without interference. Case 3 has the first l_{01} flow finishing before l_{23} , testing the case where per-link sharing changes while interference is still active.

12 Summary and Conclusions

12.1 Results Summary

12.2 Conclusions

All 104 individual test points across 9 experiments match analytical predictions within the 1% tolerance. The experiments verify the following aspects of the `ncsim` interference model and execution engine:

1. **PHY-layer baseline** (Exp. 1): The SNR-to-MCS-to-rate pipeline produces correct rates across the full distance range, including the out-of-range fallback.

Table 11: Summary of all verification experiments.

#	Experiment	Test Points	Result
1	Link Length vs. Data Rate	20	PASS
2	Parallel Link Separation	15	PASS
3	Two Transmitters to Same Receiver	18	PASS
4	Three Parallel Links (Multi-Phase)	16	PASS
5	Staggered Transfer Start	4	PASS
6	Per-Link Bandwidth Sharing	10	PASS
7	N -Way Bianchi Scaling ($N = 2 \dots 8$)	7	PASS
8	Five-Link Hidden-Terminal Cascade	5	PASS
9	Combined Sharing + Interference	9	PASS
Total		104	ALL PASS

2. **Conflict-graph contention** (Exps. 2, 3, 4, 7): The Bianchi MAC efficiency formula is correctly applied for $N = 2$ through $N = 8$ contending stations, with symmetric rate assignment.
3. **Hidden-terminal SINR degradation** (Exps. 2, 3, 4, 8): SINR computation correctly aggregates interference from multiple hidden terminals, with proper MCS re-selection and minimum-factor clamping.
4. **Expected-rate SINR model** (Exp. 4): The probability-weighted hidden-terminal model correctly produces higher throughput for outer links than middle links in the mixed regime, reflecting the physical reality that hidden terminals with contention-domain neighbors transmit less than 100% of the time.
5. **Mixed contention + hidden-terminal regimes** (Exp. 4): The model correctly composes Bianchi contention sharing with SINR degradation when some links are in the conflict graph and others are not. The throughput drop at the mixed-to-all-hidden boundary correctly reflects the loss of CSMA protection against near-neighbor interference.
6. **Dynamic recalculation** (Exps. 4, 5, 8): When transfers start or complete, all affected links are recalculated with correct partial-progress tracking (`data_remaining`), even through 4+ cascading phase transitions.
7. **Per-link fair sharing** (Exps. 6, 9): Multiple flows on the same physical link correctly receive equal bandwidth shares, with proper recalculation when flows complete.
8. **Mechanism composition** (Exp. 9): Per-link bandwidth sharing and inter-link hidden-terminal interference compose correctly, with phase transitions triggered by either mechanism.
9. **Convergence to independence** (Exp. 4): At sufficiently large separations ($s \geq 300$ m), interference power drops below the noise floor and all links achieve their solo PHY rate, confirming correct behavior in the asymptotic limit.

These results provide confidence that the `csma_bianchi` interference model in `ncsim` faithfully implements the intended analytical model and that the execution engine’s dynamic recalculation machinery is correct.

References

- [1] G. Bianchi, “Performance Analysis of the IEEE 802.11 Distributed Coordination Function,” *IEEE Journal on Selected Areas in Communications*, vol. 18, no. 3, pp. 535–547, Mar. 2000.